Confidence-Conditioned Forecasting of Market Correlation Structure as Spatiotemporal Video Prediction

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Abstract. We formulate stock market correlation dynamics as a spatiotemporal video forecasting problem: the full $N \times N$ cross-asset correlation matrix is treated as a sequence of image frames, Fiedler-reordered so that cluster structure becomes spatially coherent, and forecast with deep video architectures. Evaluating six architectures (CNN+LSTM, 3D CNN, ConvLSTM, PredRNN, a space-time Transformer, and EfficientNet transfer learning), three input modalities (return, volume, and volatility correlation), and external augmentation (India VIX, geopolitical risk, and news sentiment), we find that aggregate 5-day-ahead directional accuracy saturates at a common ≈73% ceiling regardless of model or input. Our central contribution is to show that this ceiling is a *diluted average*: directional predictability rises sharply with the magnitude of the correlation change (from 73% over all pairs to 88% on the largest moves), and *conditioning on the model's own predicted magnitude* turns the forecaster into a selective predictor that achieves 85.4% directional accuracy on its most-confident 50% of calls, 91.5% on the top quartile, and 95.6% on the top decile — and it identifies these high-conviction calls at inference time (predicted-vs-actual change magnitude correlation 0.47). We further show the resulting forecast acts as a *denoised covariance estimate*: used to infer one stock's move from a peer's, it outperforms persistence, the historical mean, and even the realized sample correlation. Finally, a systematic study shows that external regime, geopolitical, and news signals are redundant with the coupling structure already encoded in the matrix. The result is a confidence-aware correlation forecaster whose practical value lies in knowing *when* to commit.

Keywords: Stock market correlation · Spatiotemporal forecasting · Selective prediction · Confidence calibration · ConvLSTM · Covariance forecasting · Fiedler vector · Financial deep learning

1 Introduction

The pairwise correlation structure of financial assets is a fundamental quantity in risk management, portfolio construction, and systemic risk analysis. Yet while

a large body of work exists on predicting individual asset returns, relatively little attention has been paid to predicting the *future structure* of the correlation matrix itself — that is, forecasting how the entire pattern of co-movement relationships between assets will evolve over time.

Existing approaches to correlation prediction fall broadly into two camps. The first addresses pairwise correlation prediction in isolation: given the historical correlation between two specific assets, predict whether it will strengthen or weaken [1,2]. These approaches treat each pair independently and cannot capture market-wide structural shifts that simultaneously reshape many correlations. The second camp uses the correlation matrix as a static graph for downstream tasks such as return prediction or portfolio optimization [5,6], without attempting to forecast how the graph will evolve.

The gap we address is: *given the current market correlation structure, what will the full correlation matrix look like in H trading days?* This question has significant practical implications. A model that predicts future correlation structure can simultaneously forecast which asset clusters will form or dissolve, where contagion will spread in a market shock, and how portfolio diversification will change — all from a single prediction.

Our key insight is that this problem maps naturally to spatiotemporal video forecasting. At each trading day, the $N \times N$ correlation matrix constitutes a single frame, where each pixel (i, j) represents the current correlation between assets i and j. A sequence of L frames forms a video clip capturing the temporal evolution of market structure. Brightening pixels indicate strengthening correlations; fading pixels indicate weakening ones. Cluster formation appears as the emergence of bright diagonal blocks; cluster dissolution as these blocks fading.

This framing enables direct application of convolutional neural network (CNN) architectures to detect spatial cluster structure within each frame, and recurrent or three-dimensional convolutional architectures to capture how that structure evolves across frames — precisely the inductive biases needed for our prediction task.

The main contributions of this paper are:

1. A formulation of correlation matrix forecasting as spatiotemporal video prediction, with Fiedler vector reordering that transforms cluster structure into spatially coherent patterns detectable by convolutional kernels.
2. A combined structural loss — weighted MSE plus Pearson, magnitude-masked directional, and trajectory terms — that overcomes the mean-collapse failure mode of pure MSE regression on correlation matrices.
3. The central finding that the apparent ≈73% directional accuracy ceiling, which we show is shared across six architectures and three input modalities, is a *diluted average*: predictability rises monotonically with change magnitude (to 88% on the largest moves), and *confidence-conditioned selective prediction* recovers 85–96% directional accuracy on high-conviction calls that the model flags itself.

4. A demonstration that the forecast functions as a denoised covariance estimate, predicting one asset's move from a peer's better than persistence, the historical mean, and the realized sample correlation.
5. A systematic negative result: external augmentation — volume and volatility correlation, India VIX, geopolitical risk (GPR), and news sentiment (GDELT) — does not improve forecasting at this horizon, because global signals are redundant with the regime already encoded in the matrix and granular news tone carries no leading signal.

2 Related Work

2.1 Multivariate Volatility and Correlation Models

The econometric tradition models time-varying second moments through the GARCH family. Univariate GARCH [22] generalizes to the multivariate setting via the BEKK parameterization [23], the constant conditional correlation model [4], and the Dynamic Conditional Correlation (DCC) model [1], which separates volatility and correlation dynamics for tractability in higher dimensions. A parallel line exploits intraday data through realized covariance [24] and long-memory HAR models [25]. These methods are parametric and estimation-oriented: they target accurate *filtering* of the current correlation given a model structure, scale poorly to hundreds of assets, and impose strong functional-form assumptions on the dynamics. Our work instead learns the dynamics non-parametrically and forecasts the full matrix several steps ahead.

2.2 Random Matrix Theory and Correlation Networks

A large econophysics literature characterizes the *structure* of empirical correlation matrices. Random matrix theory shows that most eigenvalues of a financial correlation matrix are indistinguishable from noise, motivating spectral denoising [26,27]. Correlation-based networks — minimum spanning trees, planar maximally filtered graphs, and hierarchical clustering [14,28,15] — reveal stable community structure among assets, and studies of *correlation dynamics* [29] and behavior under market stress [30] document that this structure collapses toward market-wide co-movement during crises. This body of work informs our representation (the Fiedler ordering exploits exactly this community structure) but is descriptive: it analyzes realized matrices rather than forecasting future ones.

2.3 Graph-Based Stock Market Models

A growing deep-learning literature uses the correlation matrix to construct a stock graph and applies graph neural networks for return or movement prediction [5,6,7], including hierarchical graph attention [31] and models that fuse price graphs with textual signals [32]. Graph WaveNet [8] and related spatiotemporal GNNs predict node signals on a fixed graph. In all of these the correlation graph is a fixed *input* topology; our work treats the evolving graph itself as the prediction *target*.

2.4 Systemic Risk and Connectedness

In the risk literature, connectedness measures based on variance decompositions [33] and Granger-causal / principal-component networks [34] quantify how shocks propagate across institutions and assets. These measures are computed on realized data to monitor systemic risk; they do not provide a forward forecast of the correlation structure, which is precisely the gap our model addresses and which makes its output directly relevant to forward-looking risk monitoring.

2.5 Covariance Matrix Forecasting

Classical forecasting of the covariance matrix relies on shrinkage estimators and factor models [2,3] that improve conditioning but target estimation accuracy rather than multi-step dynamic forecasting. Recent neural approaches such as the Temporal Fusion Transformer [9] forecast low-dimensional factor or volatility representations rather than the full asset-level correlation structure. Our forecast operates on the complete 95×95 matrix and, as we show (Sec. 7.6), yields a covariance estimate that is more useful for cross-asset inference than the realized sample correlation.

2.6 Spatiotemporal and Video Deep Learning

Video understanding has produced a rich family of spatiotemporal architectures: 3D convolutional networks [10,35], the convolutional LSTM [11] and its successors PredRNN [17] and Eidetic 3D LSTM [36], and attention-based models including the Vision Transformer [37] and the space-time Transformer [18]. Transfer learning from large-scale image pretraining [12] further strengthens spatial encoders. We evaluate representatives of each family; to our knowledge none has previously been applied to financial correlation-matrix forecasting, and a central empirical finding of this paper is that they all converge to the same accuracy ceiling on this task.

2.7 Spectral Graph Theory in Finance

The Fiedler vector — the eigenvector of the second-smallest Laplacian eigenvalue — provides the optimal one-dimensional embedding of a graph and the minimum-conductance bisection [13], and spectral methods underpin much financial asset clustering [14,15]. We use the Fiedler vector not for clustering per se but for a *spatial reordering* of the correlation matrix that renders community structure as contiguous diagonal blocks, making it legible to convolutional kernels.

2.8 Selective Prediction and Confidence Calibration

Selective prediction allows a model to abstain on low-confidence inputs in exchange for higher accuracy on those it answers, with theoretical foundations

in selective classification [38] and deep-network realizations [21]; calibration of modern networks is a closely related concern [39]. We adapt this idea to correlation forecasting by using the model's own predicted change magnitude as a confidence score, tracing an accuracy–coverage curve over matrix cells — to our knowledge a novel use of selective prediction for structured financial forecasting.

2.9 External Signals in Financial Forecasting

A wide literature relates markets to external indicators: the option-implied volatility index VIX [41], text-based Economic Policy Uncertainty [40] and Geopolitical Risk [19] indices, and event/news databases such as GDELT [20]. We incorporate representatives of each — a market-implied signal (VIX), exogenous indices (GPR), and granular news tone (GDELT) — and report the (negative) result that, for correlation-structure forecasting at this horizon, they are redundant with the information already encoded in the matrix.

3 Prior Approach: Pair-Level Prediction and Its Ceiling

Before developing the correlation matrix video formulation, we investigated pair-level relationship prediction as a natural starting point. Given the historical correlation time series between two assets, can a model predict whether their relationship will strengthen, remain stable, or weaken over the next trading day?

3.1 Experimental Setup

Using the same 95 Nifty 100 stocks and 5-minute bar data, we computed three variants of the correlation score R(t) for all 4,465 unique asset pairs: a single-day rolling window (75 bars), a weekly rolling window (375 bars), and an exponentially weighted moving average ($\lambda = 0.94$). Velocity and acceleration of each score were derived via finite differences, yielding 11 features per pair per timestep. India VIX and Nifty 50 return were appended as market context, giving 12 features total.

Labels were defined as a 3-class problem (WEAKEN / STABLE / STRENGTHEN) based on the change in EWMA correlation over a 75-bar horizon, thresholded at the 33rd and 67th training percentiles.

3.2 Results and Ceiling

Table 1 reports validation accuracy for all pair-level models. All architectures converge to approximately 56% accuracy regardless of complexity.

Table 1. Pair-level relationship prediction (3-class). All models converge to the same ceiling regardless of architecture complexity.

Model	Val Accuracy	vs. Random
Random baseline	33.3%	—
XGBoost (12 features)	55.6%	+22.3pp
LSTM (12 features)	55.8%	+22.5pp
Ensemble (XGB+LSTM)	56.1%	+22.8pp
Transformer + GNN	55.5%	+22.2pp

3.3 Why the Ceiling Exists

The convergence of all architectures to ≈56% reveals a fundamental limitation of the pair-level formulation. When an external shock reshapes the market, it simultaneously affects all pair correlations. A model observing only one pair's history cannot distinguish between an idiosyncratic relationship change and a market-wide structural shift. Adding VIX as a market context feature provided no improvement, confirming that market context needs to be modeled as a spatial structure — the full correlation matrix — not as a scalar feature. This insight directly motivated the reformulation described in this paper.

4 Problem Formulation

4.1 Correlation Matrix Video

Let $\mathbf{r}_t \in \mathbb{R}^N$ denote the vector of log returns for N assets at time t. We compute the EWMA correlation matrix at each timestep with decay factor $\lambda = 0.94$, sampled at daily resolution (every 75 five-minute bars), producing a sequence of $N \times N$ correlation matrices $\{R_1, R_2, \ldots, R_T\}$ with $R_t^{(ij)} \in [-1, +1]$.

4.2 Fiedler Vector Reordering

The raw correlation matrix has no inherent spatial structure. We reorder assets using the Fiedler vector [13] — the eigenvector corresponding to the second-smallest eigenvalue of the graph Laplacian $L = D - A$, where $A_{ij} = \max(0, \bar{R}^{(ij)})$ is the mean positive correlation and D is the degree matrix. Sorting assets by Fiedler vector values places strongly correlated assets spatially adjacent, producing coherent diagonal block structure as shown in Fig. 2. The reordering is computed once and applied consistently to all frames, ensuring stable spatial positions across time.

4.3 Prediction Task

Given a lookback sequence of L reordered frames $\mathbf{X}_t = \{R_{t-L+1}, \ldots, R_t\}$, the model predicts the future correlation matrix:

$$\hat{R}_{t+H} = f_\theta(\mathbf{X}_t) \tag{1}$$

where H is the prediction horizon. We use $L = 20$ (20 trading days) and $H = 5$ (5 trading days ahead). The base regression term is a volatility-weighted mean squared error,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} w_t \cdot \frac{1}{N^2} \|\hat{R}_{t+H} - R_{t+H}\|_F^2, \tag{2}$$

where w_t is a volatility-based sample weight and $\| \cdot \|_F$ denotes the Frobenius norm. As discussed in Sec. 5.3, MSE alone induces mean-collapse, so we train with a combined structural objective that adds Pearson, directional, and trajectory terms.

5 Methodology

5.1 Fiedler Vector Reordering: From Arbitrary Matrix to Structured Image

The raw correlation matrix R_t has no inherent spatial structure — the ordering of assets along rows and columns is arbitrary, determined by listing order rather than financial relationships. Under random ordering, correlated assets are scattered across the matrix and cluster patterns appear as dispersed bright patches rather than contiguous blocks. Convolutional kernels, which detect local spatial patterns by sliding across the matrix, cannot learn meaningful features from such unstructured arrangements.

We resolve this by computing the Fiedler vector — the eigenvector corresponding to the second-smallest eigenvalue λ_2 of the graph Laplacian $L = D - A$. The adjacency matrix $A_{ij} = \max(0, \bar{R}^{(ij)})$ uses only positive mean correlations (negative correlations are not meaningful for community detection), and D is the diagonal degree matrix with $D_{ii} = \sum_j A_{ij}$.

The algebraic connectivity $\lambda_2 > 0$ (Fiedler value) measures how well-connected the graph is: higher values indicate a more cohesive market where clusters are strongly internally connected. In our dataset, $\lambda_2 = 17.57$, reflecting the high overall connectivity of the Nifty 100 index constituents.

The Fiedler vector $\mathbf{v}_2 \in \mathbb{R}^N$ provides the optimal embedding of graph nodes onto a line such that strongly connected nodes are placed close together. Formally, it solves:

$$\mathbf{v}_2 = \arg \min_{\mathbf{v} \perp \mathbf{1},\ \|\mathbf{v}\|=1} \mathbf{v}^T L \mathbf{v} = \arg \min_{\mathbf{v} \perp \mathbf{1},\ \|\mathbf{v}\|=1} \sum_{(i,j) \in E} A_{ij} (v_i - v_j)^2 \tag{3}$$

The objective penalizes placing correlated assets (A_{ij} large) far apart ($|v_i - v_j|$ large). Sorting assets by v_2 values therefore places correlated assets spatially adjacent. When this sorted ordering is applied to both rows and columns of R_t, correlation clusters appear as coherent diagonal blocks — the fundamental spatial pattern that convolutional kernels are designed to detect.

Critically, the Fiedler ordering is computed once from the mean correlation matrix $\bar{R}$ and held fixed across all frames. This ensures that the spatial position

of each asset is stable across time — just as objects do not teleport between video frames — allowing the CNN to learn that, for example, the bottom-right corner of the matrix consistently represents defensive/consumer stocks. In our experiment, the ordering naturally separated FMCG and pharmaceutical stocks (ITC, Hindustan Unilever, Nestle, Britannia, Torrent Pharma) at one extreme from healthcare and real estate stocks (Max Healthcare, Apollo Hospitals, Lodha, DLF) at the other, without any sector information being provided to the algorithm.

5.2 Correlation Matrix as a Video

With Fiedler ordering applied, the sequence of correlation matrices becomes structurally analogous to a grayscale video:

- **Each frame** is a snapshot of the full market correlation structure at one trading day, with pixel values in $[-1, +1]$ representing correlation intensity.
- **Pixel brightness** encodes correlation strength: bright green pixels ($R_{ij} \approx 1$) indicate strongly correlated pairs; dark red ($R_{ij} \approx -1$) indicate negatively correlated pairs; yellow ($R_{ij} \approx 0$) indicates uncorrelated pairs.
- **Diagonal blocks** correspond to asset clusters — groups of stocks moving together. Block brightening signals cluster strengthening; block fading signals cluster dissolution.
- **Temporal change** across frames is analogous to lighting change in a video: a market-wide correlation spike (e.g., during a crisis) brightens the entire image simultaneously, while sector rotation appears as specific blocks brightening while others fade.

This analogy is not merely aesthetic. It establishes that the same inductive biases that make CNNs effective for video — local spatial pattern detection, translation invariance, hierarchical feature composition, and temporal smoothness — are precisely the biases needed for correlation matrix forecasting. A 5×5 convolutional kernel applied to a correlation frame detects local co-movement substructures among groups of five spatially adjacent assets. A deeper kernel in the next layer detects how those substructures relate to each other. The LSTM across frames detects how these spatial patterns evolve — whether a cluster is steadily brightening (trend) or fluctuating (noise).

5.3 Combined Structural Loss

Training the video forecaster with pure weighted MSE produces a characteristic failure: the model collapses toward predicting the conditional mean (the historical-average matrix), because under squared error the conditional mean is the optimal point estimate. The result is low MSE but poor structural fidelity — predictions compressed toward the center of $[-1, +1]$ that barely beat the

mean-frame baseline. We attack this mean-collapse with a combined loss that supplements MSE with three structural terms,

$$\mathcal{L} = \alpha\,\mathcal{L}_{\text{MSE}} + \beta\,\mathcal{L}_{\text{Pearson}} + \gamma\,\mathcal{L}_{\text{dir}} + \delta\,\mathcal{L}_{\text{traj}}, \qquad (\alpha,\beta,\gamma,\delta) = (0.35, 0.25, 0.30, 0.10), \tag{4}$$

where, writing $\Delta\hat{R} = \hat{R} - R_t$ and $\Delta R = R_{t+H} - R_t$ for the predicted and realized change,

$$\mathcal{L}_{\text{Pearson}} = 1 - r(\hat{R}, R_{t+H}), \tag{5}$$

$$\mathcal{L}_{\text{dir}} = \mathbb{E}\Big[\,\text{ReLU}\Big(-\Delta\hat{R} \odot \Delta R\Big) \odot m\,\Big], \qquad m = \mathbb{1}[\,|\Delta R| > \tau\,], \tag{6}$$

$$\mathcal{L}_{\text{traj}} = \mathbb{E}\Big[\Big(|\Delta\hat{R}| - |\Delta R|\Big)^2\Big]. \tag{7}$$

The **Pearson term** rewards getting the relative ordering of correlations right, directly countering compression. The **magnitude-masked directional term** penalizes predicting the wrong sign of change; the mask m (with $\tau = 0.005$) restricts the penalty to non-trivial moves so the model is not driven by noise-level changes. The **trajectory term** matches the magnitude of predicted and actual change, discouraging both over- and under-shooting. The volatility weight on the MSE term is retained as $w_t = N\,\text{MSE}_t / \sum_{s\in\mathcal{B}} \text{MSE}_s$ with $N = |\mathcal{B}|$, concentrating the regression signal on high-information snapshots while preserving the effective learning-rate scale.

5.4 Multi-Channel Frame Representation

A single correlation value per pixel conveys only the current coupling level. We enrich each frame to C channels that summarize the local temporal state of every relationship: (1) the raw correlation, (2) its one-step change, (3) its five-step change, and (4–5) its rolling mean and standard deviation over a four-snapshot window. This gives the spatial encoder direct access to the velocity, acceleration, and local regime of each pair without relying on the recurrent module to reconstruct them. The same construction is applied identically to auxiliary modalities (volume and volatility correlation; Sec. 7.2) so that modality ablations remain controlled. Fig. 1 visualizes the five channels for a single snapshot.

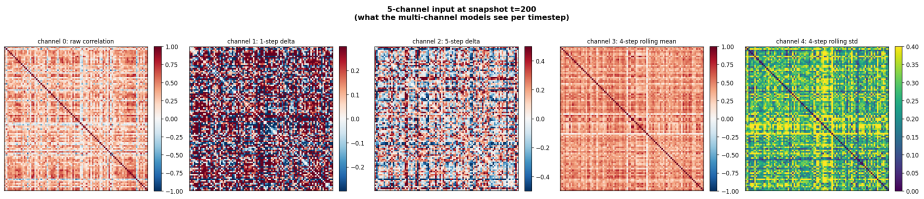


Fig. 1. The five per-pixel channels for one snapshot: raw correlation, one- and five-step change, and rolling mean and standard deviation. Each gives the encoder a different view of the local temporal state of every relationship.

5.5 Stratified Volatility Split

Standard temporal train/validation splits allocate all recent data to validation. In our dataset, the period after April 2025 is systematically less volatile than 2021–2024 (mean frame MSE 0.061 vs. 0.132). Under a naive temporal split, the validation set would consist almost entirely of stable, easy-to-predict periods, while the training set contains the volatile, hard periods. This creates a systematic evaluation bias: models would be rewarded for learning stable-period dynamics and penalized for volatile-period errors that never appear in validation.

We resolve this with stratified volatility splitting. Snapshots with MSE_t above the median (volatile) and below the median (stable) form separate strata, and within each stratum 70% are allocated to training and 30% to validation. This ensures both training and validation contain matched proportions of volatile snapshots and span the full date range (2021–2026) rather than concentrating the calm recent period in validation. The split yields 792 training and 340 validation snapshots. On this stratified validation set the persistence baseline MSE is 0.128 and the mean-frame baseline is 0.071 — the latter being the harder bar to beat, as it already captures the strong mean-reversion of correlations.

5.6 Confidence-Conditioned Selective Prediction

Aggregate directional accuracy, computed over all N^2 matrix cells, treats every prediction equally. But the vast majority of the predictable signal is concentrated in cells undergoing large coupling changes, while small changes are dominated by noise that no model can forecast (Sec. 7.3). The aggregate metric therefore *dilutes* the model's genuine skill by averaging confident, accurate predictions with unavoidable coin-flips on near-static pairs.

We make this skill usable with selective prediction [21]. The model's *predicted change magnitude* for cell (i, j), $\hat{c}_{ij} = |\hat{R}^{(ij)}_{t+H} - R^{(ij)}_t|$, is available at inference and serves as a confidence score. To predict at coverage $q \in (0, 1]$, we rank all cells by $\hat{c}_{ij}$ and emit directional predictions only for the top-q fraction, abstaining on the rest. Sweeping q traces a directional-accuracy-versus-coverage curve that exposes the accuracy the model can deliver when permitted to abstain on its low-confidence cells. As we show, this confidence score is well-calibrated: predicted and realized change magnitudes correlate at 0.47, so the cells the model is most confident about are genuinely the ones it predicts most accurately.

5.7 Cross-Stock Inference via the Forecast

A correlation forecast is ultimately useful if it improves inference about asset returns. We test this directly: given the realized return r_B of one asset over the prediction window, the forecasted coupling implies an estimate of a peer's return,

$$\hat{r}_A = \hat{\rho}_{AB} \, \frac{\sigma_A}{\sigma_B} \, r_B, \tag{8}$$

where σ_A, σ_B are trailing volatilities. This is the conditional-expectation (regression) relationship implied by the predicted correlation. It is an *inference* task — observe one leg, infer the other (as in hedging, portfolio completion, or scenario analysis) — not an ex-nihilo forecast. We compare the forecasted $\hat{\rho}$ against three baselines for ρ: persistence (last observed correlation), the historical mean correlation, and the realized future correlation (an oracle upper bound), measuring how well each predicts the actual r_A.

5.8 Model Architectures

We evaluate a broad set of video architectures, all using the same dataset, combined loss, and evaluation protocol to enable fair comparison. We describe the three that anchor the study below; EfficientNet transfer learning, PredRNN, and a factorized space-time Transformer were also evaluated (Sec. 7.2). ConvLSTM emerged as the strongest single model once multi-channel inputs were introduced and is used for all subsequent analysis.

CNN+LSTM. A 2D CNN encoder is applied independently to each frame in the lookback sequence, extracting a 256-dimensional spatial feature vector. Three convolutional layers with kernel sizes 5, 5, and 3 detect progressively larger spatial structures: the first layer detects local pair-level correlation patterns (small clusters, cluster edges), the second detects broader cluster blocks and inter-cluster relationships, and the third refines fine-grained patterns within clusters. Batch normalization and max-pooling (stride 2) after each of the first two layers reduce the spatial dimension from 95×95 to $\approx 23 \times 23$ before adaptive average pooling to a fixed 6×6 output. An LSTM with 256 hidden units and 2 layers then processes the sequence of 20 frame encodings, with the final hidden state decoded to the predicted 95×95 matrix via a three-layer MLP with Tanh output to constrain predictions to $[-1, +1]$.

The key limitation of this architecture is that spatial structure is discarded after the CNN encoder: the LSTM operates on 256-dimensional vectors, not on spatial feature maps. Temporal reasoning happens on abstract features rather than on the raw spatial structure of the correlation matrix.

3D CNN. 3D convolutional kernels operate simultaneously over spatial and temporal dimensions of the input volume $(L, N, N) = (20, 95, 95)$, treating it as a 3D tensor with a single channel. A kernel of size (T_k, H_k, W_k) detects the same spatial pattern changing over T_k consecutive frames — for example, a $(3, 3, 3)$ kernel can detect "this 3-stock cluster has been brightening over the past 3 days." This is more direct than CNN+LSTM because temporal context is captured at the level of spatial pixels rather than at the level of compressed feature vectors.

Four convolutional blocks use kernels of sizes $(3, 3, 3)$, $(3, 5, 5)$, $(5, 3, 3)$, and $(3, 3, 3)$ to capture short-range spatiotemporal patterns, broader spatial context, longer temporal patterns, and deep features respectively. The batch size was reduced to 8 due to higher GPU memory requirements (approximately 4.5GB per forward pass on the $(8, 1, 20, 95, 95)$ input tensor).

ConvLSTM [11]. Addresses the spatial discarding problem of CNN+LSTM by maintaining spatial feature maps as the hidden state throughout temporal

processing. At each timestep t, the ConvLSTM cell takes a spatial feature map $\mathbf{x}_t \in \mathbb{R}^{C\times H\times W}$ and updates spatial hidden and cell states $\mathbf{h}_t, \mathbf{c}_t \in \mathbb{R}^{C'\times H\times W}$ via convolutional gates:

$$\mathbf{i}_t = \sigma(\mathbf{W}_{xi} * \mathbf{x}_t + \mathbf{W}_{hi} * \mathbf{h}_{t-1}) \tag{9}$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_{xf} * \mathbf{x}_t + \mathbf{W}_{hf} * \mathbf{h}_{t-1}) \tag{10}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_{xc} * \mathbf{x}_t + \mathbf{W}_{hc} * \mathbf{h}_{t-1}) \tag{11}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \tag{12}$$

where $*$ denotes convolution and $\odot$ element-wise product. The spatial hidden state allows the model to track how specific regions of the correlation matrix (specific asset pairs and clusters) evolve over time without compressing spatial information to vectors. Frames are first downsampled from 95×95 to 24×24 via a convolutional encoder to reduce memory requirements.

PredRNN and space-time Transformer. To test whether more expressive temporal modeling breaks the ceiling, we additionally evaluate PredRNN [17], whose spatiotemporal LSTM cells propagate a memory state both across time and up through layers, and a factorized space-time Transformer in the style of TimeSformer [18], which patch-tokenizes each frame and applies spatial self-attention within frames followed by temporal self-attention across frames. An EfficientNet-B0 [12] transfer-learning encoder was also evaluated. None of these surpasses ConvLSTM (Sec. 7.2), reinforcing that the ceiling is a property of the data rather than of model capacity.

All models share the same decoder: a three-layer MLP ($d_{\text{hidden}} \to 1024 \to 2048 \to N^2$) with ReLU activations and Tanh output that maps the final spatial/recurrent representation to the predicted $N \times N = 95 \times 95 = 9025$ correlation values.

6 Experiments

6.1 Dataset

We use 5-minute bar data for 95 Nifty 100 constituent stocks from July 30, 2021 to April 8, 2026, comprising 87,122 aligned timesteps. EWMA correlations ($\lambda = 0.94$) are computed for all 4,465 pairs. Daily correlation matrix snapshots are built at every 75-bar interval, yielding 1,157 snapshots. The Fiedler vector naturally separates defensive/consumer stocks (FMCG, pharmaceuticals) from cyclical/growth stocks at the extremes.

Split. Stratified by volatility (Sec. 5.5): 792 training and 340 validation snapshots, with matched volatile proportions and full date-range coverage in both.

6.2 Baselines

Persistence. Predicts $\hat{R}_{t+H} = R_t$. MSE = 0.128.

Mean-frame. Predicts $\hat{R}_{t+H} = \bar{R}_{\text{train}}$. MSE = 0.071. This captures the strong mean-reversion tendency of correlations and is the harder baseline to beat.

6.3 Evaluation Metrics

MSE (unweighted). Mean squared error between predicted and actual matrices.

Pearson r. Correlation between all predicted and actual values, measuring structural agreement beyond MSE.

Directional accuracy. Fraction of pairs where predicted direction of correlation change matches actual direction.

MSE on volatile/stable snapshots. Separate MSE for snapshots above and below the volatility threshold.

6.4 Implementation Details

All models use the Adam optimizer (learning rate 10^{-3}, weight decay 10^{-4}), gradient clipping at 1.0, cosine annealing with warm restarts ($T_0 = 20$, $T_{\text{mult}} = 2$), and early stopping with patience 15. Inputs use a 20-snapshot lookback with the five-channel representation of Sec. 5.4. Training is on an NVIDIA RTX 4070 Laptop GPU (8GB VRAM) with batch size 16.

7 Results

7.1 Correlation Matrix Video Quality

Fig. 2 shows sample frames of the Fiedler-ordered correlation matrix video. Diagonal block structure is clearly visible and varies substantially across market regimes, confirming that the ordering produces spatially coherent cluster patterns suitable for convolutional processing.

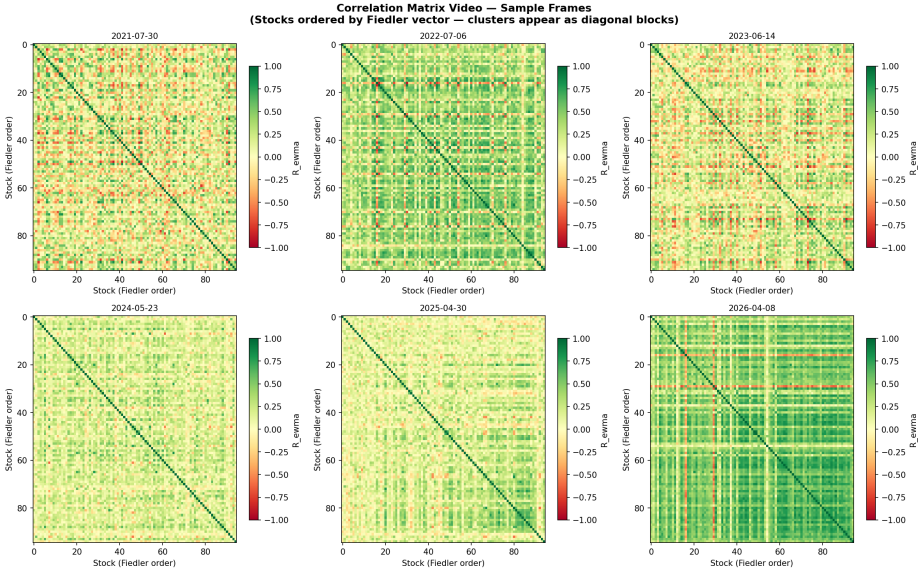


Fig. 2. Sample frames of the Fiedler-ordered correlation matrix video. Green indicates strong positive correlation; red indicates negative correlation. Visible diagonal block structure confirms spatially coherent cluster patterns.

That the pixels carry genuine financial signal is confirmed in Fig. 3: a same-sector pair (HDFCBANK–ICICIBANK) exhibits consistently high coupling, while a cross-sector pair (RELIANCE–INFY) is lower and more variable. Fig. 4 shows the market-wide mean and dispersion of coupling over time, making regime shifts — periods where the whole market couples — directly visible as a video-level brightness signal.

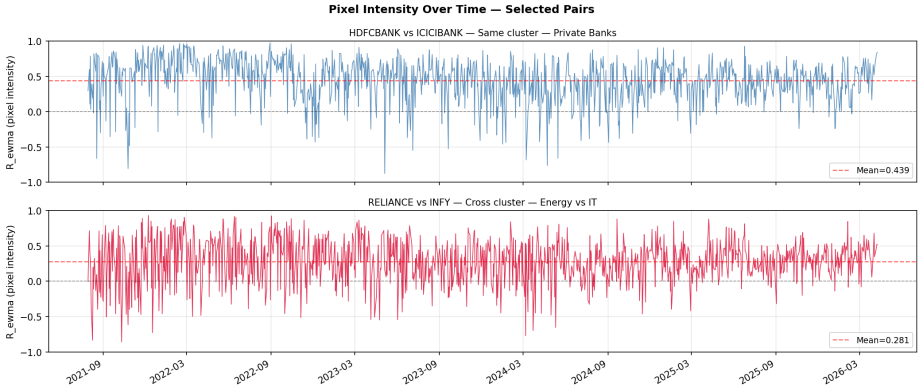


Fig. 3. Pixel intensity (EWMA correlation) over time for two asset pairs. The same-sector pair stays bright; the cross-sector pair is dimmer and more variable, confirming that pixel values carry meaningful financial signal.

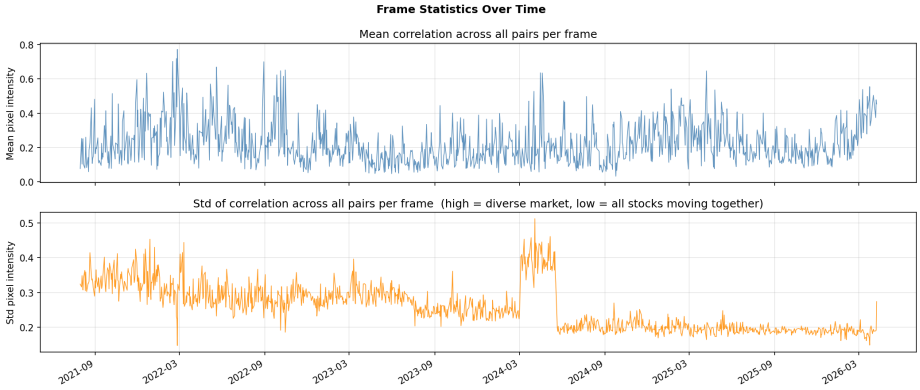


Fig. 4. Frame-level statistics over time: mean coupling (top) and cross-sectional dispersion (bottom). High mean with low dispersion marks market-wide co-movement regimes; the reverse marks idiosyncratic, well-diversified periods.

7.2 A Shared Ceiling Across Architectures and Modalities

Table 2 reports the best model under each architecture and input modality. The striking result is convergence: every configuration — four architecture families, plus volume- and volatility-correlation input modalities — lands at MSE ≈ 0.069 and directional accuracy $\approx 73\%$. PredRNN's spatiotemporal memory

and the Transformer's global attention do not surpass ConvLSTM; adding volume correlation is neutral and volatility correlation is mildly harmful (the latter is a near-global factor that dilutes the per-pair signal). The inter-architecture spread (0.0686–0.0721) is comparable to the inter-seed variance we observe on a fixed architecture, indicating that the ceiling is a property of the prediction problem, not of model capacity or input modality.

Table 2. Best configuration per architecture and input modality on the stratified validation set. All converge to MSE ≈ 0.069, directional accuracy ≈ 73%. Combined loss; multi-channel inputs.

Configuration	**MSE↓**	**Pearson r↑**	**Dir. Acc.↑**
Persistence baseline	0.128	—	50.0%
Mean-frame baseline	0.071	—	—
CNN+LSTM	**0.0681**	**0.387**	0.732
3D CNN	0.0683	0.376	**0.733**
ConvLSTM	0.0686	0.372	0.730
PredRNN	0.0721	0.295	0.724
Space-time Transformer	0.0712	0.317	0.726
ConvLSTM + volume corr.	0.0697	0.360	0.729
ConvLSTM + volatility corr.	0.0721	0.293	0.723

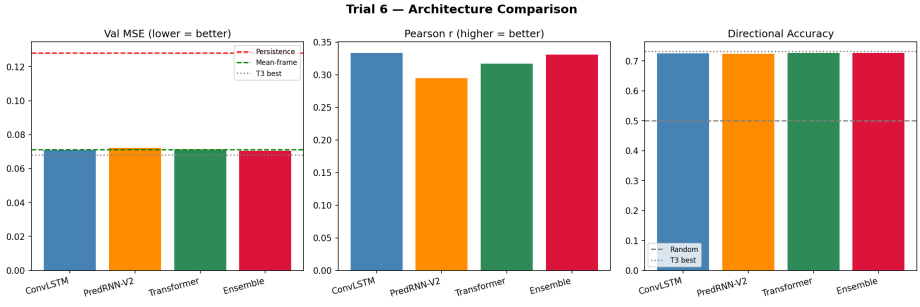


Fig. 5. Architecture comparison (ConvLSTM, PredRNN, space-time Transformer, ensemble). All cluster at MSE ≈ 0.07 and directional accuracy ≈ 0.73, illustrating the shared ceiling.

The input-modality result deserves emphasis because the volume and volatility channels are *not* redundant with returns: Fig. 6 shows that, across representative pairs, rolling volume- and volatility-correlation carry information largely

independent of return-correlation (average cross-stream correlations of 0.18 and 0.25). Fig. 7 visualizes the three modality videos side by side — they exhibit visibly different block structure. Despite this genuine independence, adding them as input channels does not help: the modality ablation ladder (Fig. 8) shows volume neutral and volatility mildly harmful relative to the return-only baseline. Independent information is not the same as *predictive* information for this target.

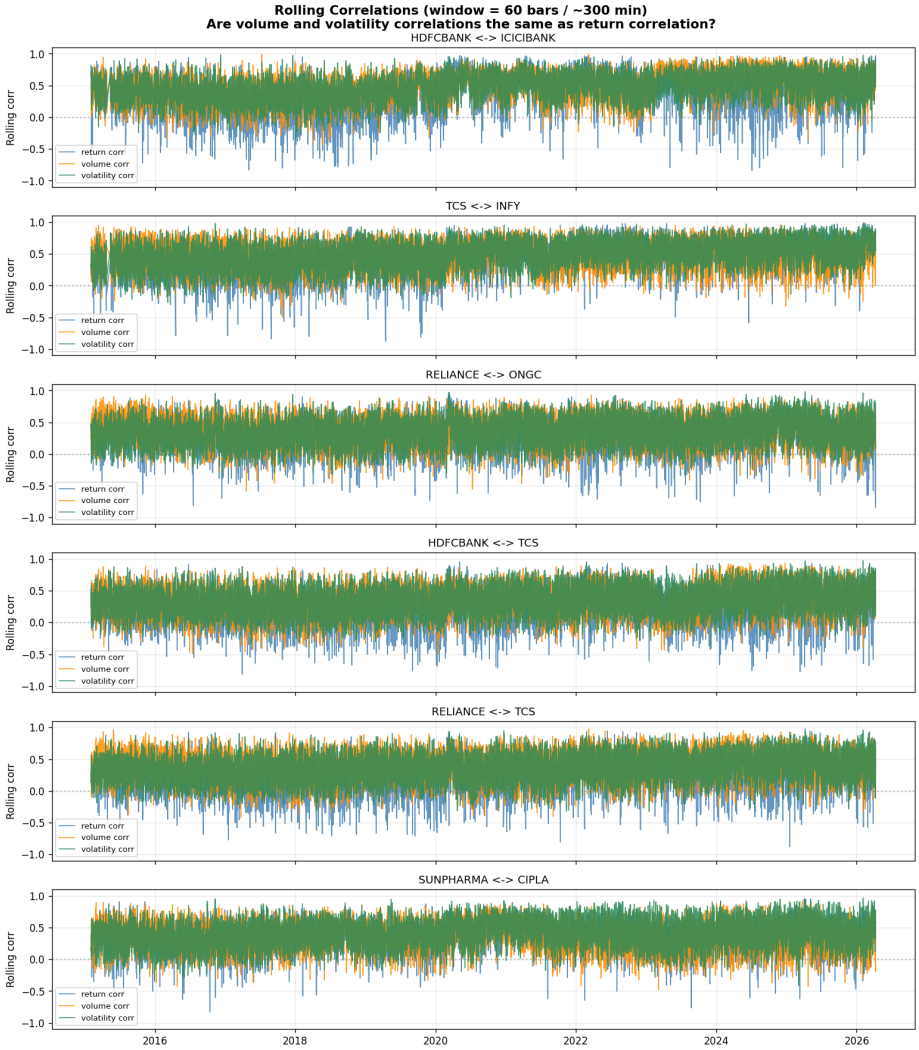


Fig. 6. Rolling return-, volume-, and volatility-correlation for representative pairs. Volume and volatility coupling move largely independently of return coupling, confirming they are distinct signals.

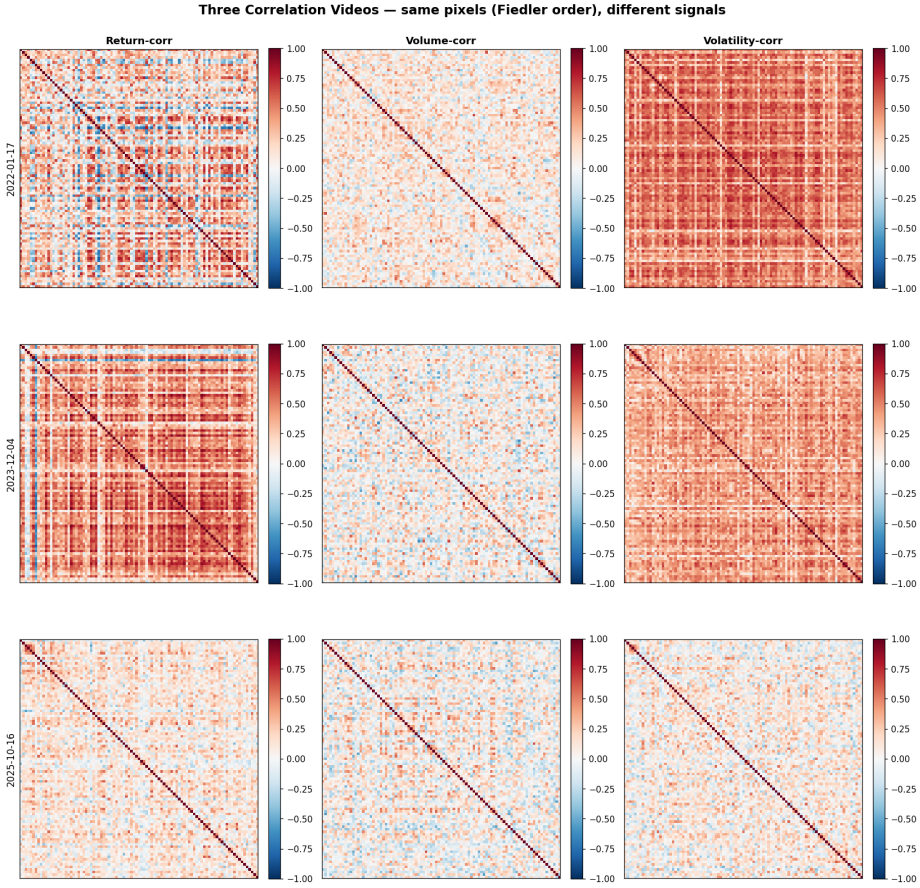


Fig. 7. The three correlation-video modalities at matched snapshots (identical Fiedler layout): return (left), volume (center), volatility (right). The distinct block patterns confirm independent structure; the per-frame cross-correlations are low.

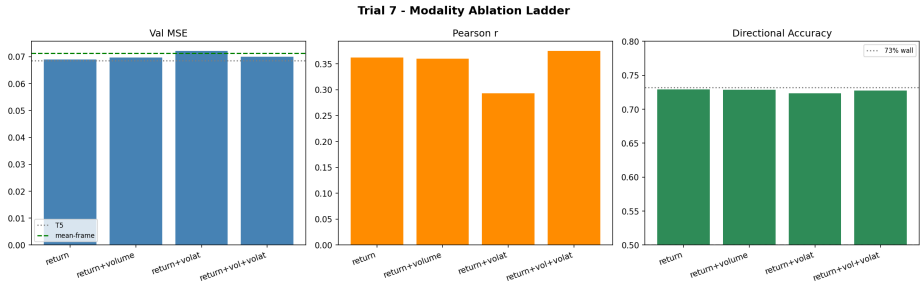


Fig. 8. Modality ablation ladder: ConvLSTM trained on return only, return+volume, return+volatility, and all three. No modality beats the return-only baseline; volatility correlation degrades it.

7.3 Predictability Rises with Change Magnitude

The aggregate 73% obscures strong structure. Table 3 stratifies directional predictability by the magnitude of the actual correlation change: it rises monotonically from 73% over all pairs to 88% on changes exceeding 0.30. A one-line mean-reversion rule (predict that coupling reverts toward its 20-snapshot mean) already attains 73% over all pairs — matching the deep models — and the same rule reaches 88% on the largest moves. The implication is twofold: (i) most of the deep model's aggregate skill coincides with mean-reversion, and (ii) the genuinely predictable signal lives in the large-magnitude changes, which the aggregate metric averages away.

Table 3. Best directional accuracy stratified by magnitude of the true correlation change. Predictability rises sharply with magnitude.

Change magnitude	% of cells	Dir. Acc.
all cells	100.0%	0.734
> 0.10	76.2%	0.790
> 0.15	65.1%	0.816
> 0.20	54.9%	0.841
> 0.30	37.5%	0.884

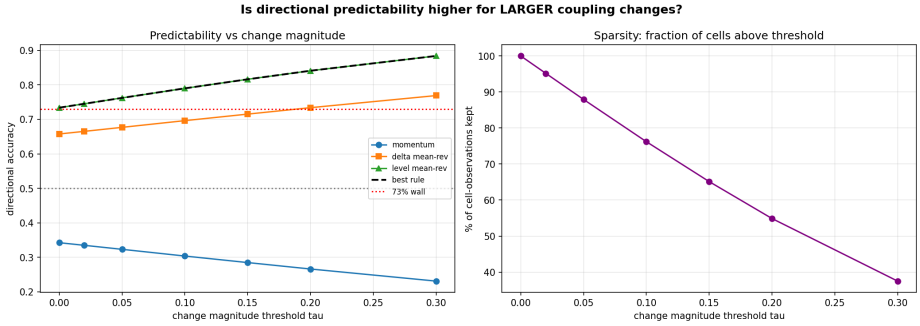


Fig. 9. Directional predictability versus change-magnitude threshold (left) and the corresponding fraction of cells retained (right). Predictability rises steeply with magnitude; the dashed line marks the 73% aggregate.

We stress that this is *not* a sparse-event regime: large changes are common (the median five-day coupling change is 0.23), and they do not concentrate on a fixed subset of "event-prone" pairs — Fig. 10 shows the per-pair frequency of large moves is nearly uniform and only weakly persistent over time. There is thus no static watchlist to exploit; the structure to exploit is the *magnitude* of each predicted change, which motivates the confidence-conditioned approach below.

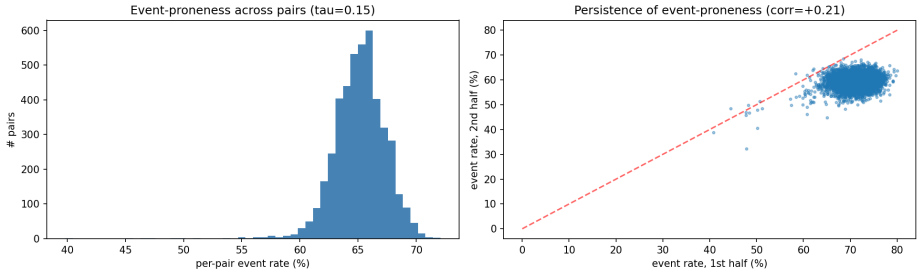


Fig. 10. Structure of large coupling changes. Left: distribution of per-pair large-move frequency, which is concentrated in a narrow band (no small set of chronically volatile pairs). Right: weak persistence of per-pair move frequency across the first vs. second half of the sample.

7.4 Selective Prediction: The Model Knows When to Commit

Because the predictable signal concentrates in large moves, and because the model's *predicted* change magnitude correlates with the true magnitude at 0.47, we can recover this skill at inference via selective prediction (Sec. 5.6). Table 4

sweeps coverage: emitting predictions only for the most-confident fraction of cells. Directional accuracy climbs from 73.7% at full coverage to 85.4% at 50% coverage, 91.5% at 25%, and 95.6% at the top decile. The model thus delivers high-accuracy directional calls precisely on the relationships it is most confident about — identified using only information available at prediction time.

Table 4. Selective prediction: directional accuracy as a function of coverage, conditioning on the model's own predicted change magnitude.

Strategy	Coverage	Dir. Acc.
Predict everything (aggregate)	100%	0.737
Top half by confidence	50%	0.854
Top quartile by confidence	25%	0.915
Highest-conviction decile	10%	0.956

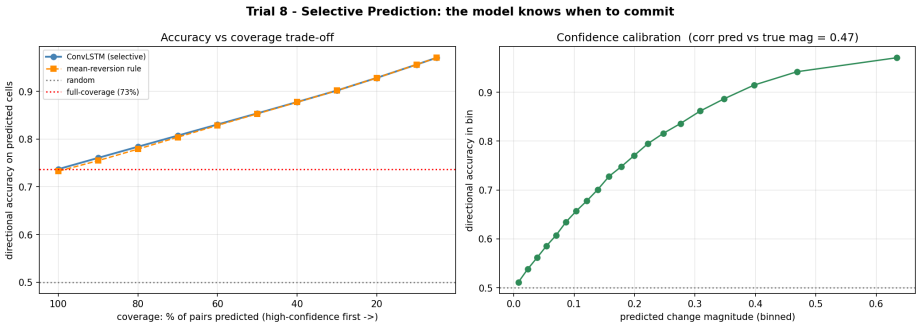


Fig. 11. Selective prediction. Left: directional accuracy versus coverage (high-confidence cells first); the model far exceeds the full-coverage 73.7% as it is allowed to abstain. Right: calibration — accuracy increases with the model's predicted change magnitude (predicted-vs-true magnitude correlation 0.47).

7.5 External Augmentation is Redundant

Table 5 summarizes a systematic study of external signals. India VIX, trained as a second stream in a controlled VIX-on vs. VIX-zeroed comparison, does not improve (and marginally degrades) the forecast. Diagnostic tests on geopolitical risk (GPR) and sector-level news sentiment (GDELT) show that global signals — including geopolitical risk [19] — carry only a weak leading component that is redundant with the regime already visible in the matrix (VIX correlates 0.46 with current average coupling), while granular news tone from GDELT [20] carries

essentially no leading signal (pooled partial correlation 0.00 with sector-coupling change). The conclusion is consistent across signal types: at this horizon the predictive content is intrinsic to the coupling structure, and external augmentation does not extend it.

Table 5. External-signal study. "Partial lead" is the partial correlation with the coupling-change residual after removing mean-reversion; the model column reports the trained two-stream outcome where applicable.

Signal	Type	Outcome
India VIX	global, market-derived	partial lead +0.22; trained model no gain
Geopolitical (GPR)	global, exogenous	partial lead +0.12; redundant with VIX
News tone (GDELT)	granular, exogenous	pooled partial lead 0.00; no signal

Fig. 12 shows the controlled two-stream comparison for VIX: across both aggregate MSE and the full selective-prediction coverage curve, the VIX-augmented model is statistically indistinguishable from (and marginally below) the identical VIX-zeroed control.

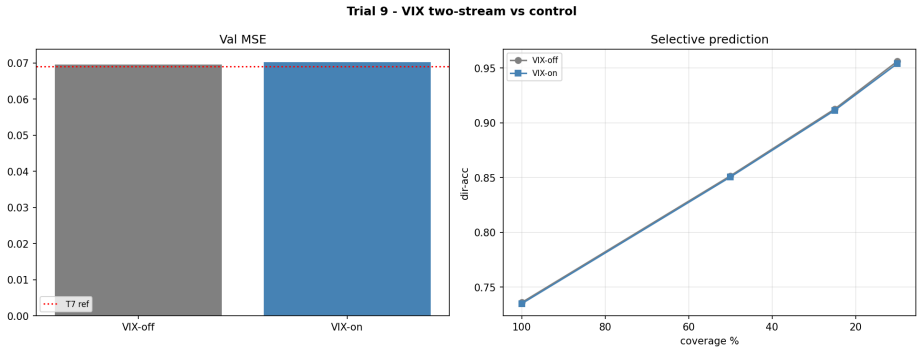


Fig. 12. Controlled VIX augmentation. Left: validation MSE, VIX-on vs. the identical VIX-zeroed control. Right: selective-prediction coverage curves for both. VIX provides no improvement, confirming that the global volatility regime is already encoded in the matrix.

7.6 Cross-Stock Inference: A Denoised Covariance

Finally, Table 6 evaluates the forecast as an inference tool: predicting one asset's return from a peer's via the implied regression (Sec. 5.7). Measured by the correlation between predicted and actual returns, the forecast (0.283) outperforms persistence (0.188), the historical mean (0.276), and — notably — the realized

sample correlation (0.236). The forecast is thus a *denoised* covariance estimate: smoother and more useful for inference than the noisy realized correlation. Its advantage over the static historical mean concentrates on the pairs where it predicts an unusual relationship (on cells where the forecast's sign disagrees with the historical norm, it scores 0.57 versus the historical rule's 0.42), consistent with the magnitude finding that the model's value lies in catching regime changes.

Table 6. Cross-stock inference: correlation between predicted and actual peer return r_A when inferred from r_B via each correlation estimate.

Correlation estimate ρ	**corr(predicted r_A, actual r_A)**
Persistence (last observed)	0.188
Realized (oracle, true future)	0.236
Historical mean	0.276
Forecast (ours)	**0.283**

8 Discussion

8.1 Why the Pair-Level Ceiling Exists: A Structural Argument

The convergence of all pair-level architectures to ≈56% accuracy, regardless of model complexity, deserves careful analysis because it reveals something fundamental about the structure of the prediction problem rather than about the models themselves.

Consider a pair of stocks (i, j) whose correlation at time t is $R_t^{(ij)}$. Whether this correlation strengthens or weakens over the next H bars is determined by two distinct processes: (1) the *idiosyncratic* relationship between stocks i and j specifically, and (2) the *market-wide* structural shift that simultaneously affects many correlations. A pair-level model observing only $R_t^{(ij)}$ cannot distinguish between these two processes. When India VIX spikes and all correlations rise simultaneously, a pair-level model sees this as "strengthening" in the pair's relationship. But the pair has no special relationship with VIX — it is simply caught in a market-wide tide.

Formally, let $R_t^{(ij)} = S_t^{(ij)} + \epsilon_t^{(ij)}$ where $S_t^{(ij)}$ is the systematic component driven by market-wide factors and $\epsilon_t^{(ij)}$ is the idiosyncratic component. A pair-level model can observe $R_t^{(ij)}$ but cannot decompose it into S and ϵ without observing the full matrix R_t. The full matrix provides the context needed to determine whether the current correlation level is high relative to the *whole market* (suggesting a market-wide event) or high only relative to *this specific pair's history* (suggesting an idiosyncratic change). This is precisely why the video-based formulation — which provides the full $N \times N$ matrix as context — breaks through the ceiling.

8.2 Why Fiedler Ordering Enables Convolutional Learning

The Fiedler vector ordering is not merely a visualization technique — it is a functional prerequisite for convolutional processing of the correlation matrix. Without it, convolutional kernels cannot extract meaningful features because spatially adjacent cells have no systematic relationship.

To see this concretely: a 3×3 kernel applied at position $(20, 20)$ of a randomly ordered correlation matrix sees the correlations between three arbitrary, unrelated stocks. The same kernel applied at the same position of a Fiedler-ordered matrix sees the correlations within a subgroup of stocks that the spectral ordering has identified as belonging to the same community. The latter is a meaningful financial quantity (local cluster cohesion); the former is noise.

The Fiedler ordering exploits a deep connection between spectral graph theory and community detection. The Fiedler vector partitions the graph at the highest-conductance cut — the partition that minimizes inter-community edges relative to intra-community edges. When assets are ordered by Fiedler values, assets with similar Fiedler values are in the same community (cluster), and their pairwise correlations appear as a bright block along the diagonal. The "brightness" of the block measures community cohesion.

This also explains why the ordering can be computed on the mean correlation matrix rather than requiring snapshot-by-snapshot recomputation. The Fiedler ordering provides the *canonical* arrangement of assets in correlation space — the stable background ordering against which temporal fluctuations (blocks brightening or fading) become legible as a video signal.

8.3 The 73% Ceiling is a Diluted Average

The most important interpretive point of this work is that the apparent ceiling is an artifact of how directional accuracy is aggregated. Computed over all N^2 cells, the metric averages two populations: a large majority of near-static pairs whose tiny changes are dominated by noise, and a minority of pairs undergoing large, structurally-driven moves. The first population is essentially unforecastable and pins the average near the mean-reversion rate; the second is where genuine skill lives (Table 3). The convergence of every architecture and modality to 73% is therefore not a failure to extract signal — it is the signature of an average dominated by an unforecastable majority. Once predictions are conditioned on magnitude, the underlying skill (85–96% on confident calls) becomes visible. The practical lesson is that correlation forecasters should be deployed and reported with a coverage axis, not a single aggregate number.

8.4 Why Confidence Conditioning Works

Selective prediction is only useful if the model can identify its own reliable predictions *ex ante*. It can: the predicted change magnitude correlates with the realized magnitude at 0.47, so ranking cells by predicted magnitude approximately ranks them by accuracy. This calibration arises naturally from the combined

loss — the trajectory and directional terms encourage the model to commit to a sized change only when the input dynamics support it, and to predict near-zero change otherwise. The result is a forecaster that abstains, in effect, on the pairs where it would otherwise be guessing, and the accuracy–coverage curve (Fig. 11) quantifies the trade-off a practitioner can choose along.

8.5 Why External Signals are Redundant

That neither market-derived (VIX) nor exogenous (GPR, GDELT) external signals help is initially surprising but has a clean explanation. Global risk signals are highly correlated with the *current* level of average market coupling (VIX correlates 0.46 with it), and that level is already an input to the model — it is literally visible in the correlation frame. A single scalar therefore adds little beyond what the matrix encodes, and cannot differentially reshape the 4,465 pairwise relationships. Granular news tone, while genuinely independent of VIX, carries no measurable leading signal for sector coupling — consistent with news being largely priced by the time it is published. The boundary this establishes is itself a contribution: it locates the predictive content firmly in the endogenous coupling structure and its dynamics, not in external context.

8.6 The Honest Role of Mean-Reversion

We are deliberate about what the model does and does not add over a trivial baseline. A one-line rule — coupling reverts toward its recent mean — already attains the 73% aggregate and rises to 88% on large moves, so the bulk of the model's directional skill coincides with mean-reversion. The deep model's distinctive contributions are narrower but real: (i) a calibrated confidence signal that enables selective prediction, and (ii) a denoised covariance estimate that beats the realized sample correlation for cross-stock inference (Table 6), with its edge over the static mean concentrated on regime-change pairs. We regard honest delineation of this boundary as more valuable than an inflated headline number.

Fig. 13 grounds this in the predictability structure of the data: frame-to-frame coupling changes are strongly mean-reverting at one step (delta autocorrelation ≈ -0.47) and decorrelate almost completely beyond two steps, while the mean-frame baseline MSE is nearly flat across horizons. The predictable signal is therefore short-horizon and mean-reversion-dominated — precisely the regime in which a calibrated, confidence-aware forecaster is the appropriate tool.

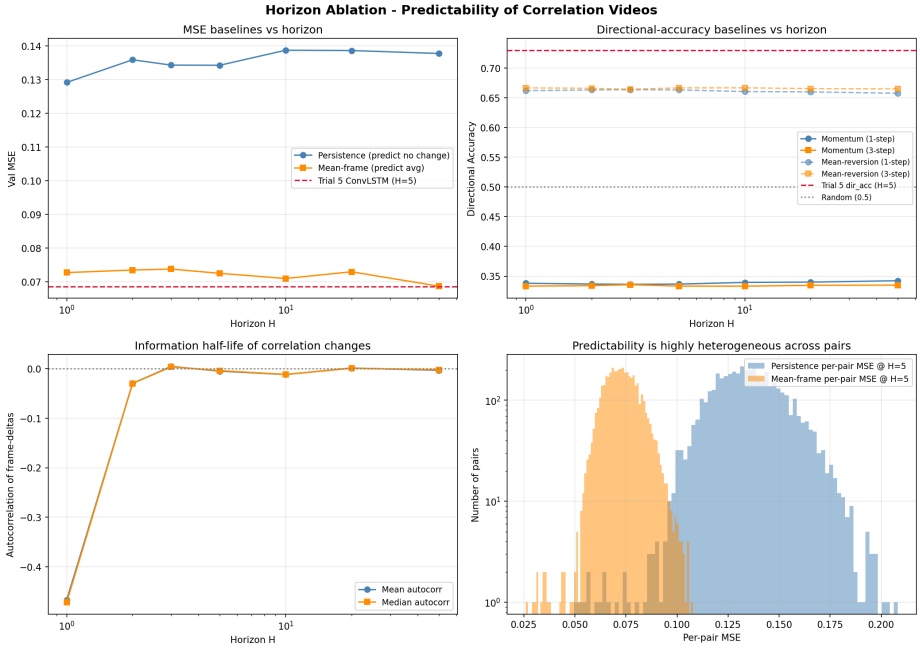


Fig. 13. Predictability structure versus horizon (no training). Baseline MSE and directional accuracy across horizons, the autocorrelation decay of frame-to-frame changes, and the heterogeneous per-pair error distribution. Signal is concentrated at short horizons and dominated by mean-reversion.

8.7 Practical Applications

The predicted future correlation matrix has several direct applications in finance, each exploiting a different aspect of the model's output.

Diversification monitoring. Portfolio managers construct portfolios assuming diversification based on current correlations. If the model predicts that previously uncorrelated assets will become highly correlated over the next 5 trading days, the effective diversification of the portfolio will decrease — meaning the portfolio carries more concentrated risk than its current construction implies. The model enables proactive rebalancing before the correlation shift materializes, rather than reactive adjustment after it has already affected portfolio performance.

Forward-looking stress testing. Standard Value-at-Risk computations use the current covariance matrix. If the model predicts a regime of elevated correlations (as in the 2022 high-correlation period), risk managers can compute VaR under the predicted future correlation structure, providing a forward-looking measure of tail risk. Under selective deployment, elevated-coupling warn-

ings issued only at high confidence are correct 85–96% of the time (Table 4), rather than the 73% implied by the undifferentiated aggregate.

Pairs trading signal generation. Statistical arbitrage strategies [16] profit from the convergence of highly correlated pairs that have temporarily diverged. Selective prediction is a natural fit: acting only on the model's high-confidence coupling-shift calls yields directional accuracy in the high 80s to mid 90s, and the cross-stock-inference result (Sec. 7.6) shows the forecast translates into better estimates of one leg's move given the other — directly useful for constructing and hedging the trade.

Cluster-based sector rotation. The predicted future matrix can be passed to the same Louvain community detection algorithm used in this work to produce a predicted future cluster structure. Changes in predicted cluster membership indicate sector rotation: stocks predicted to leave one cluster and join another are candidates for momentum-following or mean-reversion trades depending on the historical behavior of similar transitions.

8.8 Limitations and Future Directions

Modest edge over mean-reversion. As emphasized throughout, a one-line mean-reversion rule attains the same aggregate directional accuracy as the deep models. The model's value is the calibrated confidence enabling selective prediction and its role as a denoised covariance estimate, not a large gain in raw point accuracy. We view stating this boundary plainly as a strength, but it tempers any claim of a categorical improvement over simple baselines.

EWMA smoothing confound. The correlation video is built from EWMA correlations, which are autocorrelated and mean-reverting by construction. Part of the observed mean-reversion may therefore reflect the smoothing filter rather than market dynamics. Confirming that the magnitude and selective-prediction findings persist under raw windowed-correlation inputs (which we compute but do not analyze here) is important future work.

Single-peer cross-stock inference. The cross-stock result infers a return from a *single* peer. Using the full predicted matrix to infer a stock jointly from all peers — a factor-style completion — should be substantially stronger and is a natural extension.

External data was tested, not omitted. Unlike many studies, we explicitly evaluated external augmentation (volume and volatility correlation, India VIX, geopolitical risk, news sentiment) and found it redundant at this horizon. The one external avenue we did not exhaust is granular, scheduled, per-asset information — notably earnings calendars — which (unlike fuzzy news tone) is a hard leading fact and could in principle reshape specific pairs. Longer/shorter horizons are also open: our diagnostics indicate the predictable signal concentrates at the one-step horizon, where the confident-prediction ceiling may be higher.

End-to-end portfolio optimization. The natural economic validation is to feed the forecast covariance into a minimum-variance portfolio and measure

realized out-of-sample volatility against sample-covariance and persistence baselines; the cross-stock result suggests this would be favorable. Connecting the model to a differentiable portfolio layer and training for a risk-adjusted return objective directly is a promising direction.

9 Conclusion

We formulated correlation matrix forecasting as spatiotemporal video prediction and, across six architectures and three input modalities, found a shared ≈73% directional-accuracy ceiling. Our central result is that this ceiling is a *diluted average*: predictability rises with the magnitude of the correlation change (to 88% on the largest moves), and confidence-conditioned selective prediction recovers 85.4%, 91.5%, and 95.6% directional accuracy at 50%, 25%, and 10% coverage — on high-conviction calls the model identifies itself (predicted-vs-actual magnitude correlation 0.47). The forecast further acts as a denoised covariance estimate, predicting one asset's move from a peer's better than persistence, the historical mean, and the realized sample correlation. A systematic study shows that external regime, geopolitical, and news signals are redundant with the coupling structure already encoded in the matrix.

The resulting system is a confidence-aware correlation forecaster whose practical value lies in knowing *when* to commit: rather than a single undifferentiated prediction, it offers an accuracy–coverage trade-off that a risk manager or arbitrageur can choose along. We are explicit that the bulk of the aggregate directional skill coincides with mean-reversion; the model's distinctive contributions are calibrated confidence and denoised cross-stock inference. Future work includes a formal minimum-variance portfolio backtest, a one-step-horizon study where the predictable signal is strongest, joint multi-peer inference from the full predicted matrix, and granular scheduled external signals such as earnings calendars.

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